User-ready Tether Anchorages for Restraint Systems and Booster Seats (Standard 210.1)

- **[SOR/2013-117, s. 7]**

Application

210.1 (1) Subject to subsection (2), this section applies to every

(a) passenger car;

(b) three-wheeled vehicle;

(c) multi-purpose passenger vehicle and truck with a GVWR of 3 856 kg or less and an unloaded vehicle weight of 2 495 kg or less;

(d) school bus; and

(e) bus, other than a school bus, with a GVWR of 4 536 kg or less.

(2) This section does not apply to

(a) a designated seating position at which a built-in restraint system is provided that is not part of a removable vehicle seat; or

(b) a hearse.

General

(3) Subject to subsections (3.2) and (3.3), a user-ready tether anchorage shall be installed in a vehicle, other than a convertible or an open-body type vehicle,

(a) in the case of a vehicle, other than a school bus, that has only one row of forward-facing designated seating positions, at all forward-facing designated seating positions other than that of the driver;

(b) in the case of a vehicle, other than a school bus, that has two or more rows of designated seating positions, and not more than three forward-facing designated seating positions located to the rear of the first row of designated seating positions, at all forward-facing designated seating positions located to the rear of the first row of designated seating positions;

(c) in the case of a vehicle, other than a school bus, that has two or more rows of designated seating positions, and four or more forward-facing designated seating positions located to the rear of the first row of designated seating positions, at three forward-facing designated seating positions located to the rear of the first row of designated seating positions, with at least one user-ready tether anchorage being installed at a forward-facing designated seating position in the second row of designated seating positions, and at least one user-ready tether anchorage being installed at a forward-facing inboard designated seating position, if such a designated seating position exists;

(d) in the case of a school bus that has not more than 24 passenger designated seating positions, and only one forward-facing designated seating position other than that of the driver and those of a bench seat that is next to an emergency exit and contains an adjacent seat, at that forward-facing designated seating position;

(e) in the case of a school bus that has not more than 24 passenger designated seating positions, and two or more forward-facing designated seating positions other than that of the driver and those of a bench seat that is next to an emergency exit and contains an adjacent seat, at two of those forward-facing designated seating positions;

(f) in the case of a school bus that has 25 or more, but not more than 65, passenger designated seating positions, and not more than three forward-facing designated seating positions other than that of the driver and those of a bench seat that is next to an emergency exit and contains an adjacent seat, at all of those forward-facing designated seating positions;

(g) in the case of a school bus that has 25 or more, but not more than 65, passenger designated seating positions, and four or more forward-facing designated seating positions other than that of the driver and those of a bench seat that is next to an emergency exit and contains an adjacent seat, at four of those forward-facing designated seating positions;

(h) in the case of a school bus that has 66 or more passenger designated seating positions, and not more than seven forward-facing designated seating positions other than that of the driver and those of a bench seat that is next to an emergency exit and contains an adjacent seat, at all of those forward-facing designated seating positions; and

(i) in the case of a school bus that has 66 or more passenger designated seating positions, and eight or more forward-facing designated seating positions other than that of the driver and those of a bench seat that is next to an emergency exit and contains an adjacent seat, at eight of those forward-facing designated seating positions.

(3.1) A user-ready tether anchorage shall be available for use at all times, except when the designated seating position at which it is installed is not available for use because the vehicle seat has been removed or converted to an alternate use, such as the carrying of cargo.

(3.2) If a lower universal anchorage system is installed at a passenger designated seating position in the first row of designated seating positions in accordance with subsection 210.2(8), one user-ready tether anchorage shall be installed at that designated seating position.

(3.3) The number of user-ready tether anchorages required under paragraphs (3)(b) and (c) may be reduced by one if a user-ready tether anchorage is installed in the first row of designated seatings positions in accordance with subsection (3.2).

(3.4) [Repealed, SOR/2013-117, s. 8]

(4) The portion of a user-ready tether anchorage that is designed to bind with the tether strap hook shall be readily accessible and, if under a cover, the cover shall be identified by one of the symbols or the mirror image of one of the symbols set out in Figure 2 and shall be removable without the use of tools.

Tether Anchorage Positioning

(5) Subject to subsections (5.1) and (7), the portion of each user-ready tether anchorage that is designed to bind with a tether strap hook shall be located within the shaded zone, as shown in Figures 3 to 7, of the designated seating position for which it is installed, with reference to the H-point of a template described in section 4.1 of SAE Standard J826, Devices for Use in Defining and Measuring Vehicle Seating Accommodation (July 1995), if

(a) the H-Point of the template is located

(i) at the unique Design H-Point of the designated seating position, as defined in section 3.11.1 of SAE Recommended Practice J1100, Motor Vehicle Dimensions (February 2001), at the full downward and full rearward position of the seat, or

(ii) in the case of a designated seating position that is equipped with a lower universal anchorage system, midway between the two lower universal anchorage system bars;

(b) the torso line of the template is at the same angle to the vertical plane as the vehicle seat back with the seat adjusted to its full rearward and full downward position and the seat back in its most upright position; and

(c) the template is positioned in the vertical longitudinal plane that contains the H-point of the template.

(5.1) In the case of a bus, no portion of the user-ready tether anchorage shall be located on the bus floor.

(6) [Repealed, SOR/2008-72, s. 8]

(7) The portion of a user-ready tether anchorage in a vehicle that is designed to bind with the tether strap hook may be located outside the shaded zone referred to in subsection (5), if no part of the shaded zone is accessible without removing a seating component of the vehicle and the vehicle is equipped with a routing device that

(a) ensures that the tether strap functions as if the portion of the anchorage designed to bind with the tether strap hook were located within the shaded zone;

(b) is at least 65 mm behind the torso line, in the case of a non-rigid-webbing-type routing device or a deployable routing device, or at least 100 mm behind the torso line, in the case of a fixed rigid routing device; and

(c) when tested after being installed as it is intended to be used, is of sufficient strength to withstand, with the user-ready tether anchorage, the force referred to in subsection (8).

Strength Requirements

(8) Subject to subsection (10), every user-ready tether anchorage in a row of designated seating positions shall, when tested, withstand the application of a force of 10 000 N

(a) applied by means of one of the following types of test devices, installed as a child restraint system would be in accordance with the vehicle manufacturer's installation instructions, namely,

(i) a test device shown in Figures 12 to 16, or

(ii) a test device shown in Figures 7 and 8 of section 210.2, in the case of a designated seating position having a lower universal anchorage system;

(b) applied

(i) in a forward direction parallel to the vehicle's vertical longitudinal plane through the X point on the test device, and

(ii) initially, along a line above the horizontal line, at an angle of 10° ± 5° to it.

(c) attained within 30 seconds, at any onset force rate of not more than 135 000 N/s; and

(d) maintained at a 10 000-N level for a minimum of one second.

(9) [Repealed, SOR/2008-72, s. 8]

(10) If the zones in which user-ready tether anchorages are located overlap and if, in the overlap area, a user-ready tether anchorage is installed that is designed to accept the tether strap hooks of two restraint systems or booster seats simultaneously, each portion of the user-ready tether anchorage that is designed to bind with a tether strap hook shall withstand the force referred to in subsection (8) when it is applied to both portions simultaneously.

(11) If a bench seat in a bus or a row of designated seating positions in another vehicle has more than one user-ready tether anchorage and a distance of 400 mm or more, measured in accordance with Figure 20, separates the midpoints of adjacent designated seating positions, the force referred to in subsection (8) or (10), as the case may be, shall be applied simultaneously to each user-ready tether anchorage in the manner specified in that subsection.

(12) The strength requirement tests shall be conducted with the vehicle seat adjusted to its full rearward and full downward position and the seat back in its most upright position.

Figure 1 [Repealed, SOR/2013-117, s. 8]

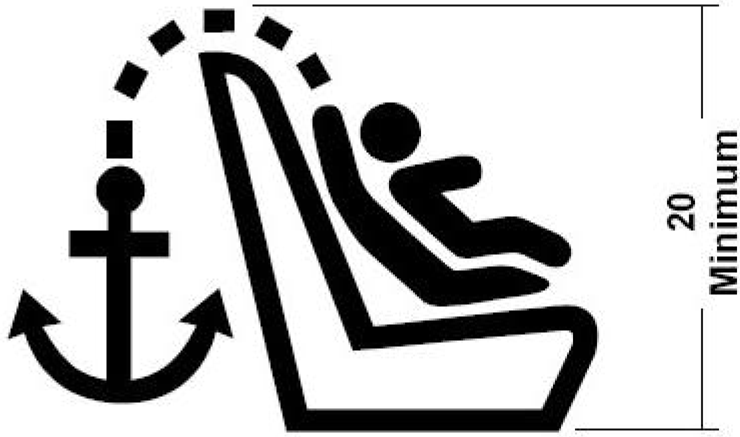

Figure 2 — Symbol Used to Identify the Location of a User-ready Tether Anchorage That Is under a Cover

Notes

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Dimensions in mm

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Drawing not to scale

3

Symbol may be embossed

4

Colour of the symbol is at the option of the manufacturer

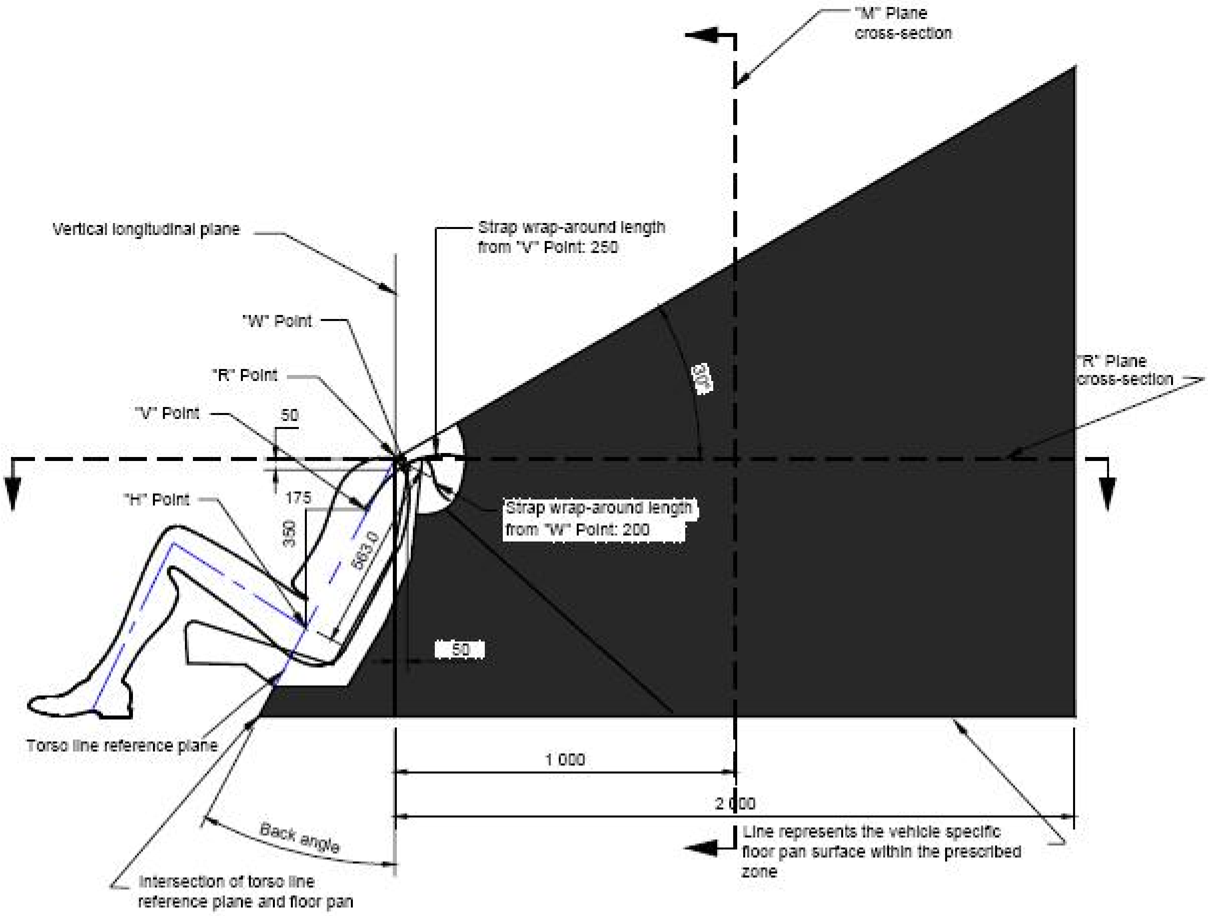

Figure 3 — Side View, User-ready Tether Anchorage Location

Notes

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Dimensions in mm, except where otherwise indicated

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Portion of user-ready tether anchorage that is designed to bind with the tether strap hook to be located within sh
zone

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Drawing not to scale

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4

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"R" Point: Shoulder reference point

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5

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"V" Point: V-reference point, 350 mm vertically above and 175 mm horizontally back from H-point

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6

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"W" Point: W-reference point, 50 mm vertically below and 50 mm horizontally back from "R" Point

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7

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"M" Plane: M-reference plane, 1 000 mm horizontally back from "R" Point

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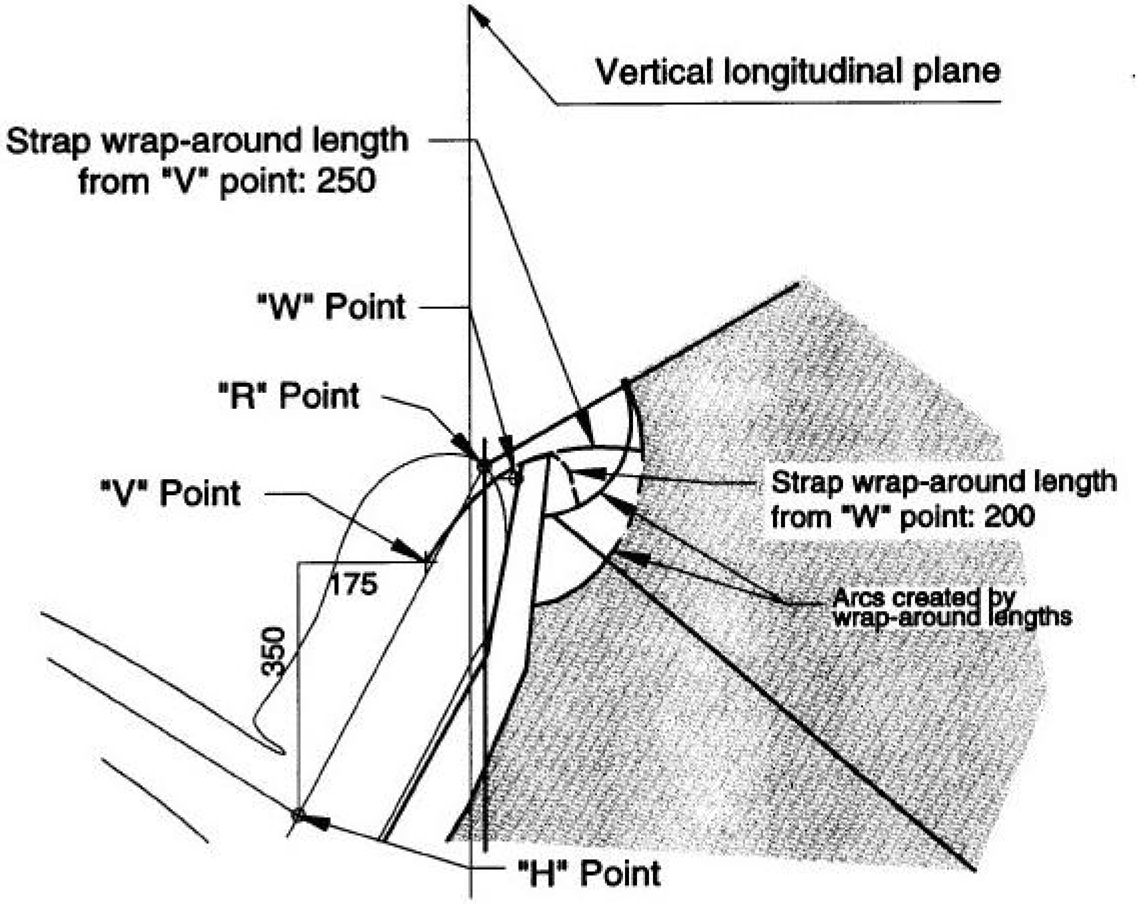

Figure 4 — Enlarged Side View of Strap Wrap-around Area, User-ready Tether Anchorage Location

Notes

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Dimensions in mm, except where otherwise indicated

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2

Portion of user-ready tether anchorage that is designed to bind with the tether strap hook to be located within sh
zone

3

Drawing not to scale

4

"R" Point: Shoulder reference point

5

"V": V-reference point, 350 mm vertically above and 175 mm horizontally back from H-point

6

"W" Point: W-reference point, 50 mm vertically below and 50 mm horizontally back from "R" Point

7

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  “M” Plane: M-reference plane, 1 000 mm horizontally back from “R” Point

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Figure 5 — Plan View (R-plane Cross Section), User-ready Tether Anchorage Location

Notes

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  Dimensions in mm, except where otherwise indicated

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  2

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  Portion of user-ready tether anchorage that is designed to bind with the tether strap hook to be located within sh zone

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-

  3

-

  Drawing not to scale

-
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4

•

"R" Point: Shoulder reference point

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5

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"V" Point: V-reference point, 350 mm vertically above and 175 mm horizontally back from H-point

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6

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"W" Point: W-reference point, 50 mm vertically below and 50 mm horizontally back from "R" Point

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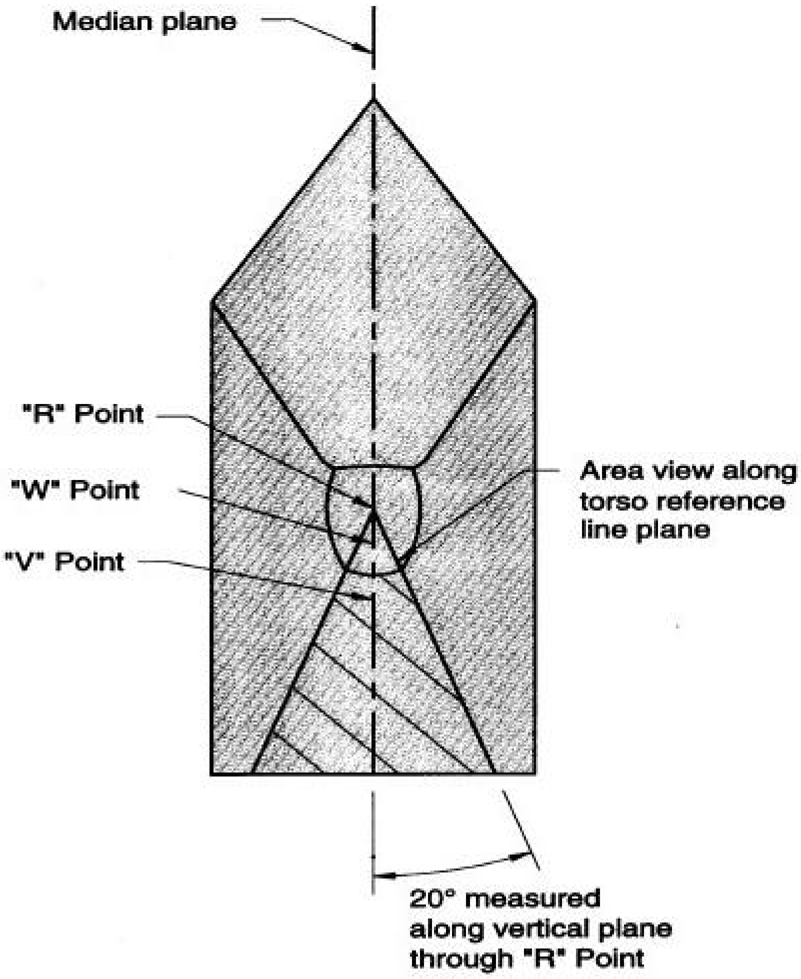

Figure 6 — Front View, User-ready Tether Anchorage Location

Notes

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Portion of user-ready tether anchorage that is designed to bind with the tether strap hook to be located within sh zone

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Drawing not to scale

3

"R" Point: Shoulder reference point

4

"V" Point: V-reference point, 350 mm vertically above and 175 mm horizontally back from H-point

5

"W" Point: W-reference point, 50 mm vertically below and 50 mm horizontally back from "R" Point

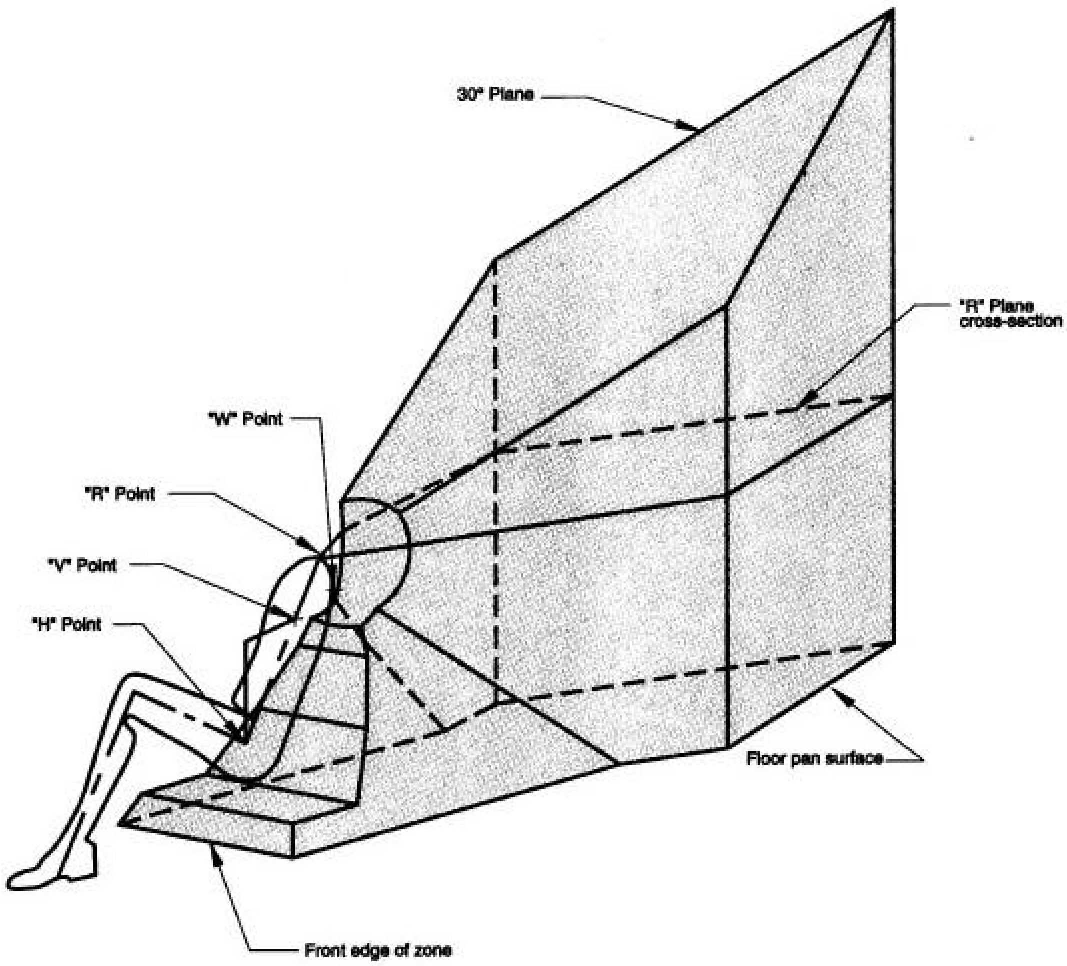

Figure 7 — Three-dimensional Schematic View of User-ready Tether Anchorage Location

Notes

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Portion of user-ready tether anchorage that is designed to bind with the tether strap hook to be located within shaded zone

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Drawing not to scale

3

“R” Point: Shoulder reference point

4

“V” Point: V-reference point, 350 mm vertically above and 175 mm horizontally back from H-point

5

“W” Point: W-reference point, 50 mm vertically below and 50 mm horizontally back from “R” Point

Figures 8 to 11 [Repealed, SOR/2008-72, s. 8]

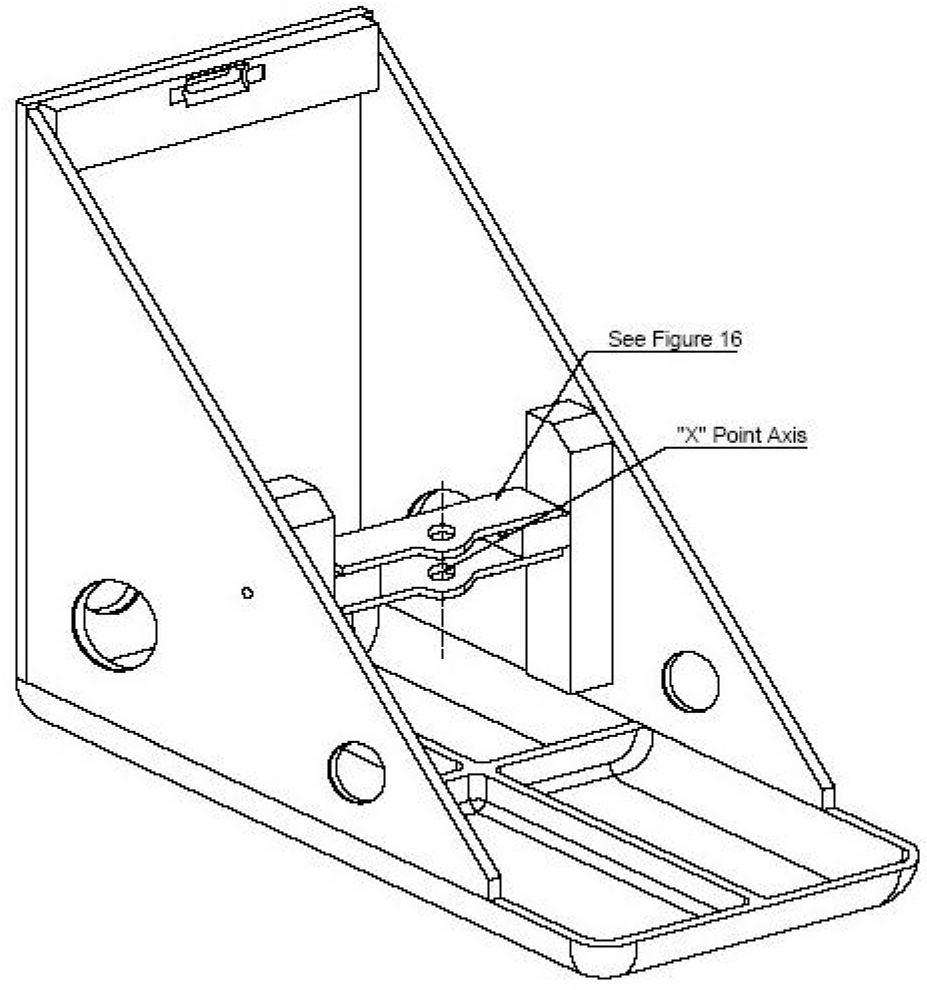

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Figure 12 — Three-dimensional Schematic View of the Static Force Application Test Device

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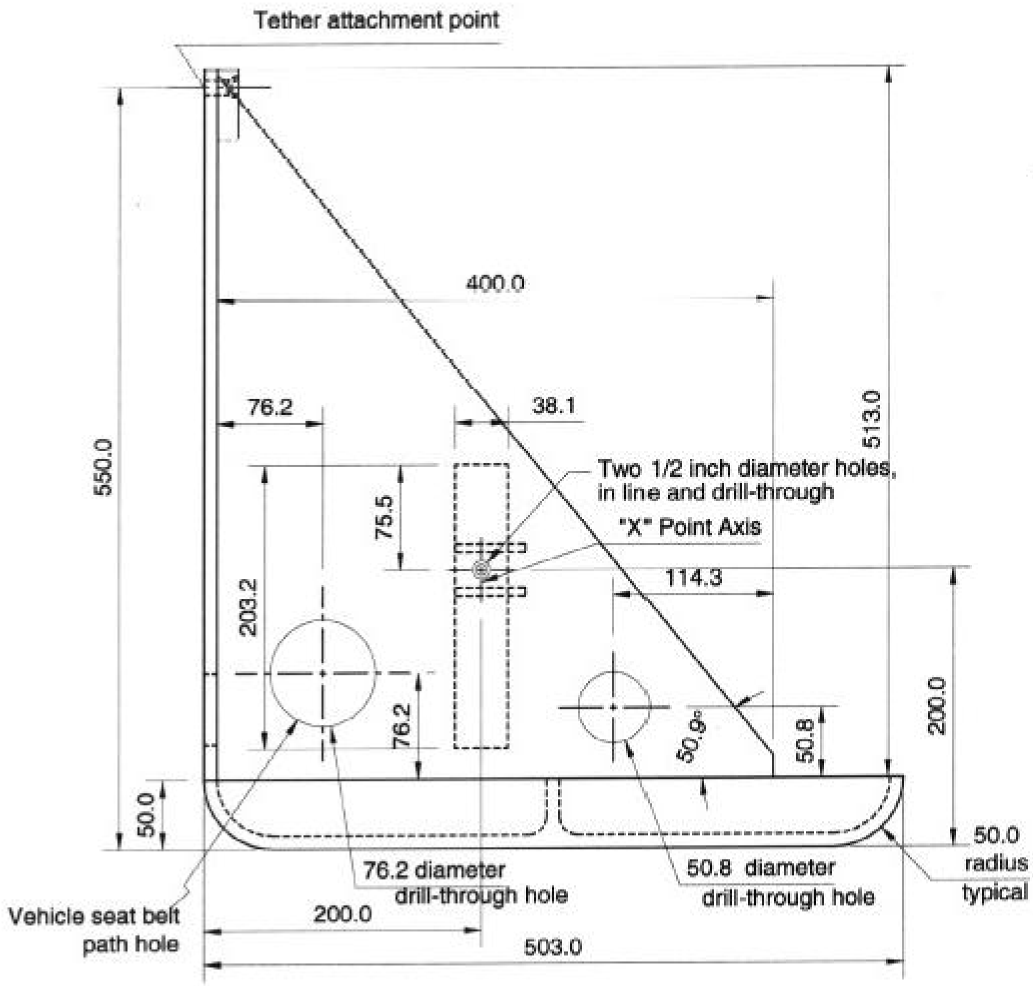

Figure 13 — Side View, Static Force Application Test Device

Notes

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Material: 6061-T6-910 Aluminum

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Dimensions in mm, except where otherwise indicated

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Drawing not to scale

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4

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Break all outside corners

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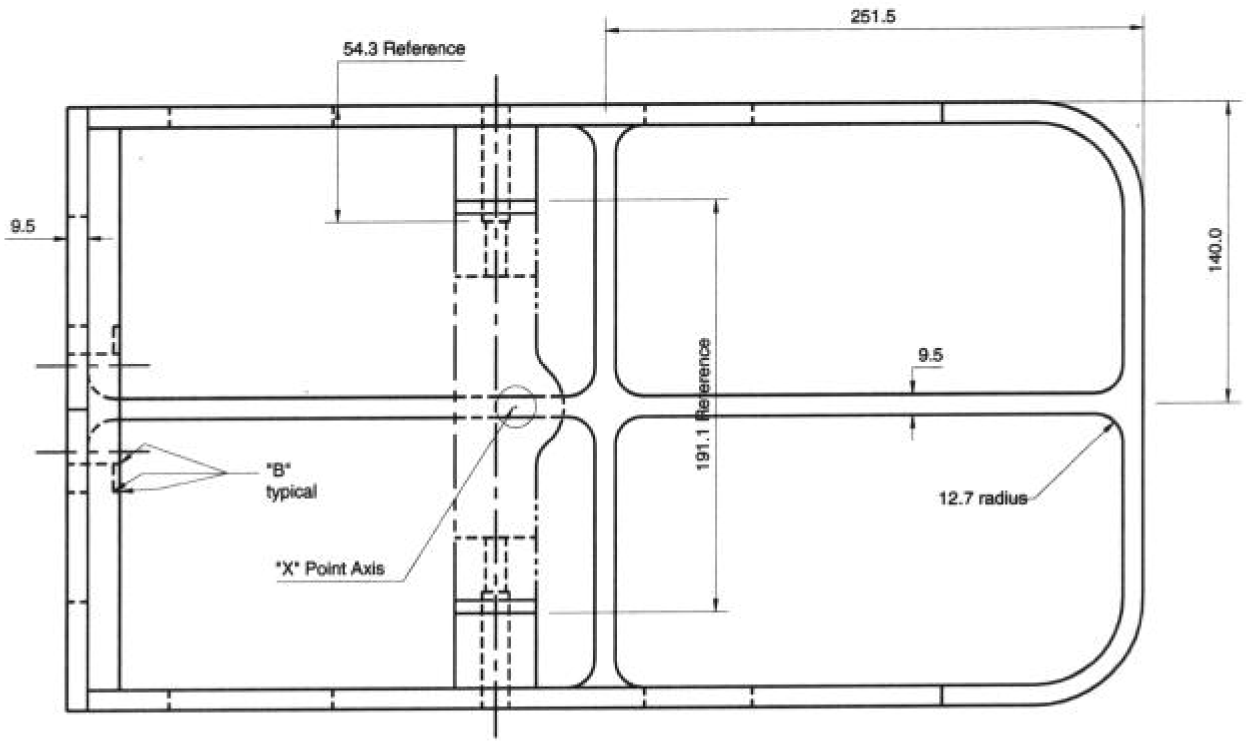

Figure 14 — Plan View, Static Force Application Test Device

Notes

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Material: 6061-T6-910 Aluminum

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Dimensions in mm, except where otherwise indicated

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Drawing not to scale

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Break all outside corners and lightning hole edges approximately 1.5 mm

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Break edges of vehicle seat belt path holes at least 44 mm

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6

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“B” = approximately 0.8 mm

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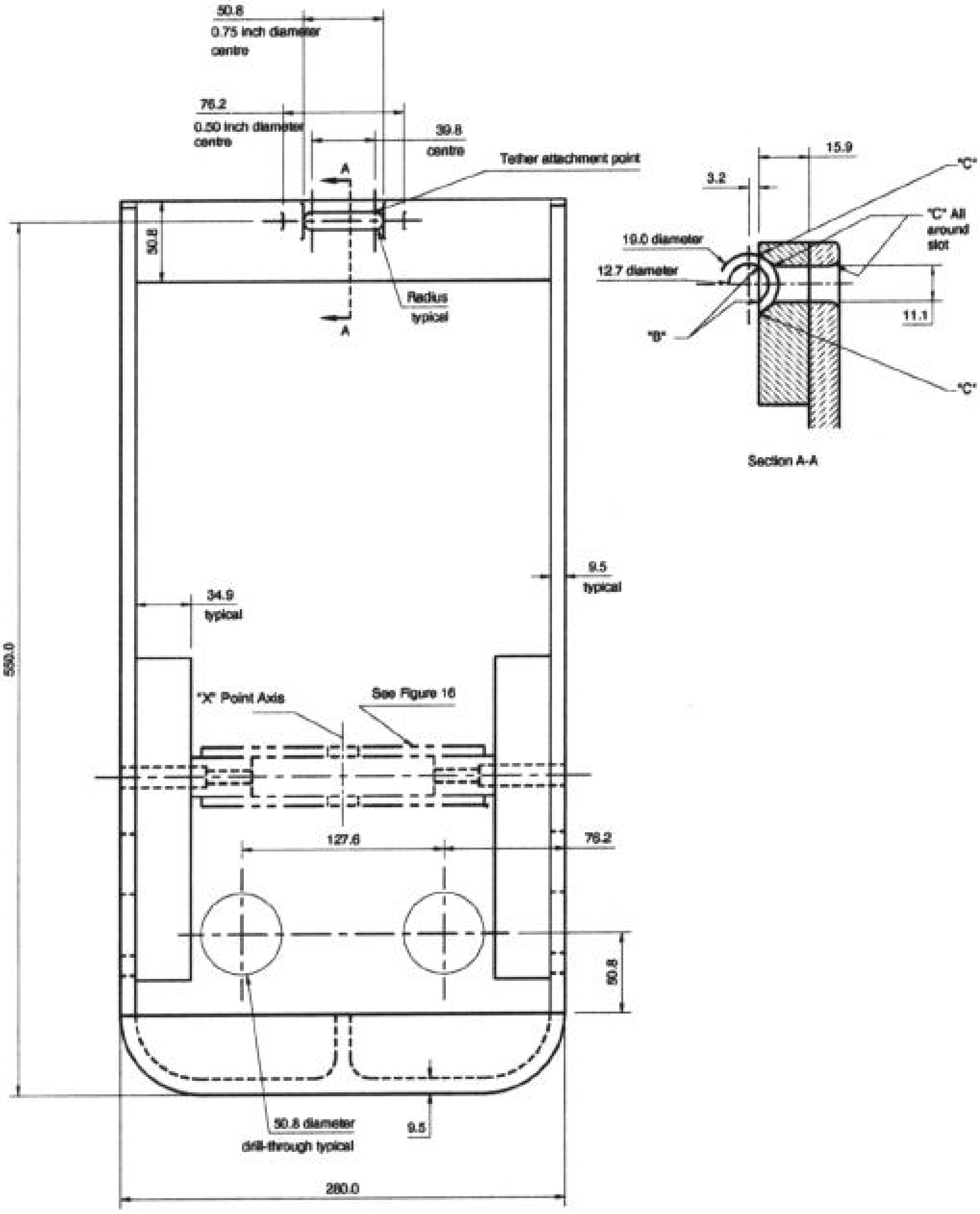

Figure 15 — Front View, Static Force Application Test Device

Notes

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Material: 6061-T6-910 Aluminum

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Dimensions in mm, except where otherwise indicated

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3

-

Drawing not to scale

-
-

4

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“B” = approximately 0.8 mm

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-

5

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“C” = approximately 3.2 mm

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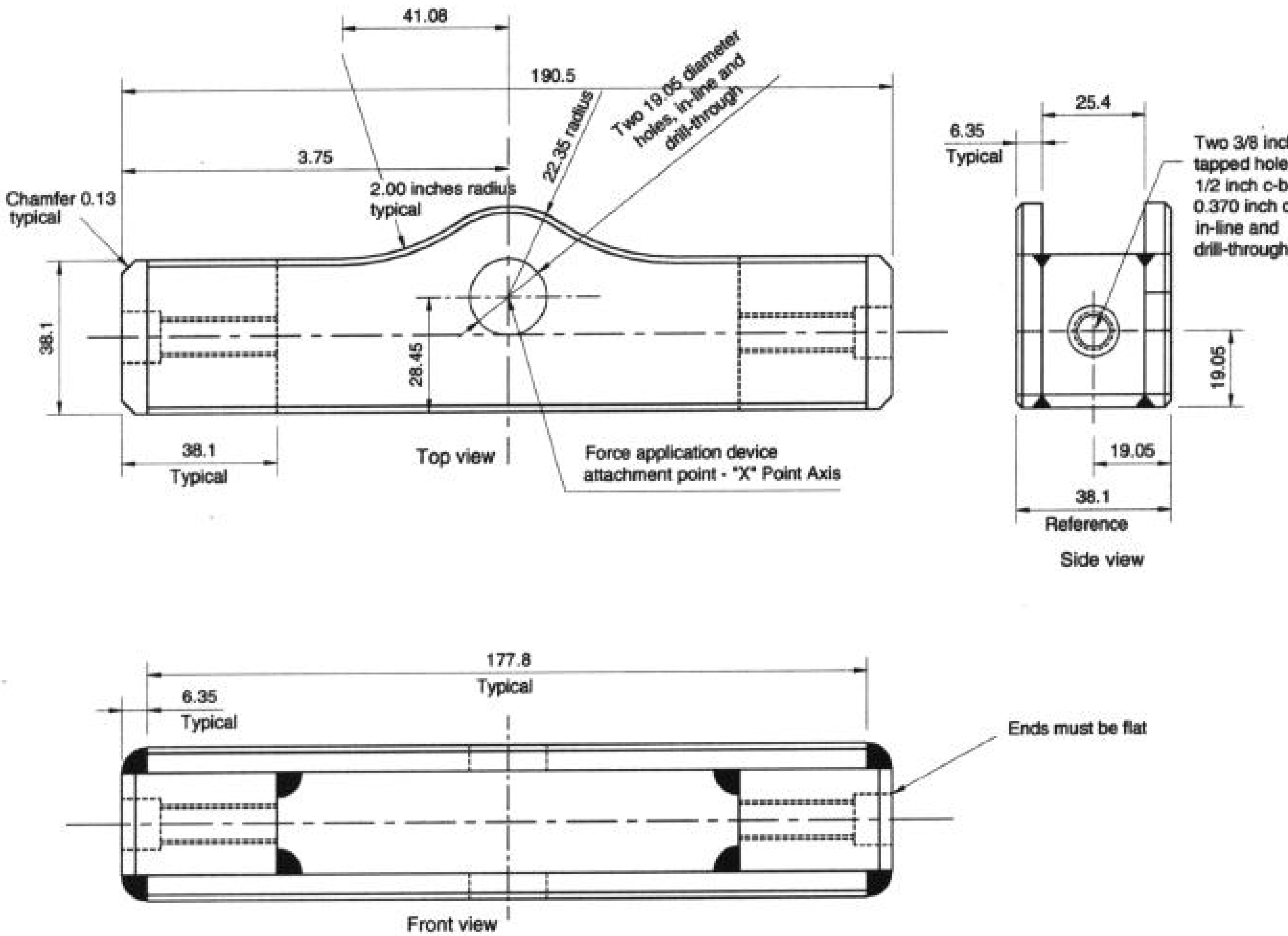

Figure 16 — Cross Bar, Static Force Application Test Device

Notes

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Material: Steel

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2

Dimensions in mm, except where otherwise indicated

3

Drawing not to scale

4

Break all outside corners approximately 1.5 mm

5

Surfaces and edges are not to be machined unless otherwise specified for tolerance

6

Saw-cut or stock size material whenever possible

- 7
- Construction to be securely welded
-

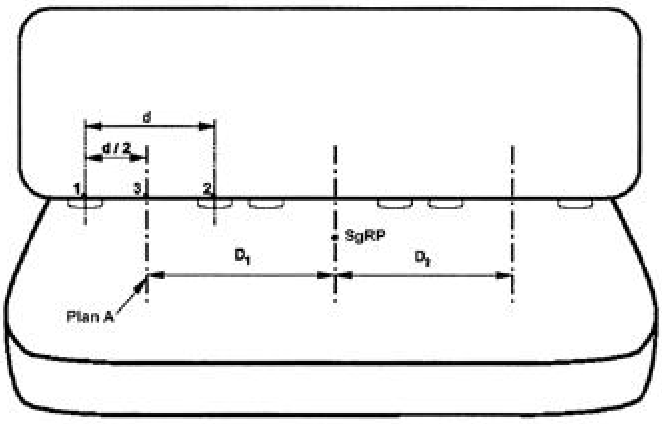
- Figures 17 and 18 [Repealed, SOR/2002-205, s. 3]
- Figure 19 [Repealed, SOR/2008-72, s. 8]
- 

Figure 20 — Measurement of Distance Between Adjacent Designated Seating Positions for Use in Simultan Testing

Legend:

- **d =**
- **centre to centre distance between the bars of a lower universal anchorage system for a given seating posit (nominal distance of 280 mm)**

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D =

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distance between vertical longitudinal planes located midway between the bars of a lower universal ancho system for a given seating position

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-

SgRP =

-

seating reference point

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Notes:

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1

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Drawing not to scale

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2

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The midpoint of a designated seating position lies in the vertical longitudinal plane that is equidistant fro the vertical longitudinal planes through the geometric centre of each of the two bars of the lower universa anchorage system installed at the seating position. For those designated seating positions that do not have lower universal anchorage system bars, the midpoint of a designated seating position lies in the vertical longitudinal plane that passes through the SgRP of the seating position

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The distance shall be measured between the vertical longitudinal planes passing through the midpoints of adjacent designated seating positions along a line perpendicular to the plane

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